# VENKATA SAI GANESH MANDA

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**Target Roles:** SDE | AI Engineer | Robotics SDE (Summer 2027 onwards)

## SUMMARY

Master's student in Robotics and Autonomous Systems (AI) at Arizona State University (GPA 4.0) with hands-on experience building scalable distributed systems, LLM-powered multi-agent pipelines, production RAG applications, and robotics software integrating computer vision, motion planning, and autonomous control. Modified vLLM source, wrote automated tests, and authored technical documentation. Proficient in Python, C++, SQL, R, OOP, data structures and algorithms, CI/CD, and Agile SDLC.

## EDUCATION

**M.S. Robotics and Autonomous Systems (AI), Arizona State University** Aug 2025 – May 2027
SCAI, Ira A. Fulton Schools of Engineering, Tempe, AZ | GPA: 4.0/4.0 | IEEE Eta Kappa Nu (HKN), Epsilon Beta Chapter
Relevant Coursework: Applied Linear Algebra, Robotic Systems I, Agentic AI, Statistical Machine Learning, Data Processing at Scale

**B.Tech. Artificial Intelligence and Data Science, IIITDM Kurnool** Sep 2020 – Jun 2024
Exchange student, University of Agder (UiA), Grimstad, Norway | Jan 2024 – Jun 2024

## TECHNICAL SKILLS AND CERTIFICATIONS

**Languages:** Python, C++, C, CUDA, R, MATLAB, SQL, JavaScript/TypeScript (basic)
**Software Eng:** OOP, Data Structures and Algorithms, Distributed Systems, Design Patterns, AI-Assisted Development (Claude Code), Agile/Scrum, CI/CD, Git (CLI), Code Review, Debugging, SDLC, TCP/IP Networking, Linux
**AI/ML:** PyTorch, TensorFlow, Scikit-learn, LangChain, RAG, Multi-Agent Architectures, OpenAI API, DeepSeek API, Gemini API, Prompt Engineering
**Robotics/CV:** ROS 2, OpenCV, Computer Vision, Homography, Perspective Transforms, Affine Calibration, State Estimation (Kalman Filtering), Motion Planning, Path Planning
**Data:** EDA, Statistical Modeling, Data Wrangling, Predictive Modeling, A/B Testing, Causal Inference, Data Visualization, NumPy, Pandas, Hadoop, Spark
**Testing and QA:** Unit Testing, Smoke Testing, Pytest, Functional Testing, Test Automation, Defect Tracking, Validation and Verification
**Cloud/Infra:** AWS (ECS Fargate, ALB, S3, IAM), Azure AI Studio, ChromaDB, REST APIs, FastAPI, SLURM, HPC, GitHub Actions, Docker, Kubernetes, MLOps, Microservices
**Certifications:** ML Specialization (Coursera/Stanford); Big Data and Spark (EdX)

## PROJECTS

**CodeLens: RAG-Based Codebase Intelligence API | Independent Project** May 2026 – Present
- Shipped production RAG API (hybrid BM25 + semantic retrieval, cross-encoder reranking, ChromaDB), deployed via Docker on AWS ECS Fargate.
- Built GitHub Actions CI/CD gating merges on quality checks; cut full repo reindex from 3.5 hours to minutes.
- Cut eval cost 96% to $0.62/run (saving ~87K GPT-4o tokens/run) while holding ~80% faithfulness and answer-relevancy. https://github.com/mvsg2/CodeLens | Demo: https://youtu.be/zm7Cu2Lo5Ww

**Tau-Bench: Multi-Agent LLM Infrastructure | Arizona State University** Jan 2026 – May 2026
- Modified 11 vLLM source files to enable distributed model serving on an HPC cluster across 825 Tau-Bench runs in the airline and retail domains (165 tasks x 5 trials).
- Wrote Pytest smoke/unit tests for critic-agent eval; co-authored 115 SLURM scripts for fault-tolerant distributed runs across Qwen3 models (4B-32B).
- Root-caused 3,677 Tau-Bench agent failures; redesigned architecture around Planner-Critic-Executor to fix the dominant failure mode. https://github.com/mvsg2/tau-bench-CSE598-repo

**Autonomous Maze-Solving Robot | Arizona State University** Aug 2025 – Dec 2025
- Built Python perception-planning-control pipeline (OpenCV) converting raw camera frames into structured occupancy grids.
- Used Gemini 2.5 Flash tool calls for entrance/exit detection plus optimization-based planning for minimum-length waypoint sequences.
- Achieved 100% success across all tested 4x4 maze configs via 6-parameter affine least-squares pixel-to-robot calibration, regardless of camera orientation. https://github.com/mvsg2/RAS-545-Midterm-Project | Demo: https://www.youtube.com/watch?v=QzS8yIfglXg

**LLM-Controlled Robotic Manipulation | Arizona State University** Aug 2025 – Dec 2025
- Architected OOP agentic pipeline (Gemini API + ROS 2 + OpenCV) letting a robot arm do pick-and-place from natural language commands.
- Built Python tool-calling interface for pixel-to-world transforms with clean separation across vision, planning, and execution.
- Validated end-to-end autonomy with zero manual intervention across all evaluated object placements; published open-source with docs. https://github.com/mvsg2/RAS-545-Final-Project | Demo: https://youtu.be/sKL3z0AOqHU

## PROFESSIONAL EXPERIENCE

**ULC TA/Grader: Engineering | Arizona State University, Learning Enterprise** Aug 2026 – Present
- Served as one of two Teaching Assistants for RAS 210: Computer-Aided Design and Manufacturing, an ASU course with 110+ enrolled learners.
- Evaluated 700+ CAD/CAM projects, homework assignments, and discussion board submissions across 14 sequential modules.
- Held daily office hours for 20 hours per week, addressing learner inquiries on course material and grading.

**AI Engineering Student Intern | Semine AS, Kristiansand, Norway** Jan 2024 – Jun 2024
- Helped build an Azure document-intelligence pipeline (OCR + REST field extraction) that cut manual processing time 30% in production; benchmarked the team's approach against Azure Document Intelligence Studio.
- Debugged data-quality defects via structured EDA over 300+ invoice records in a SQL Server-backed Python codebase.
- Supported fine-tuning of document models across 200+ foreign-language invoice types, helping lift extraction accuracy 15% over baseline OCR.